Digital Signal Processing of Cortical and Muscular Dynamics

111261019王星皓

OVERVIEW

- Introduction
- Data collection
- Data preprocessing
- Segmentation of transitional stages
- Muscle synergy
- Results
- Conclusion
- References

INTRODUCTION

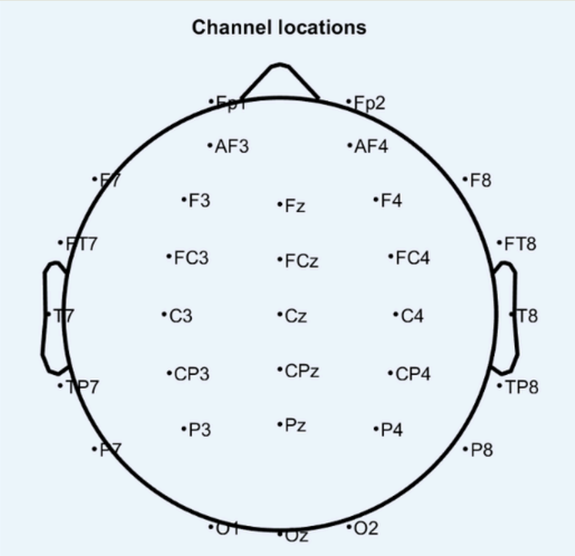
Walking is a routine yet complex task that requires coordination between the brain and muscles. To simplify motor control, the central nervous system (CNS) organizes muscle activity into synergies—co-activated groups of muscles driven by descending cortical commands. This synergy-based control is particularly critical during transitional stages—Sit-to-Stand, Stand-to-Walk, and Walk-to-Stop—when the brain must rapidly adapt postural and movement strategies. Investigating both cortical (EEG) and muscular (synergy) dynamics during these transitions can reveal how the CNS manages stability and mobility in dynamically changing conditions.

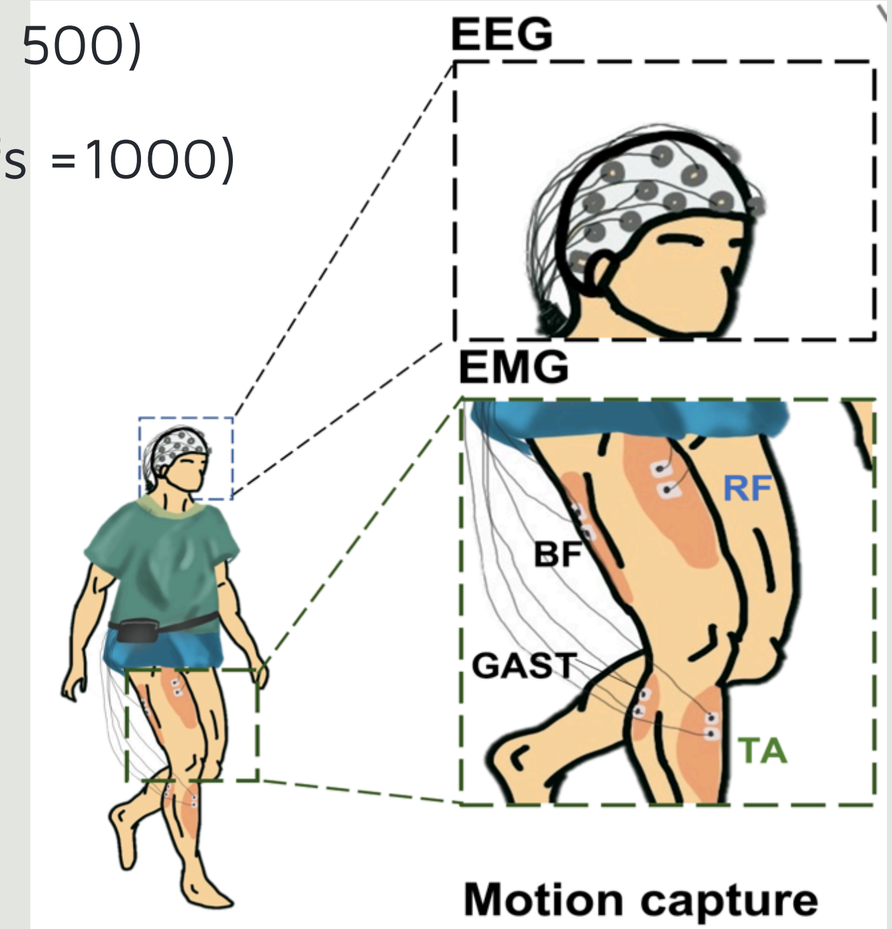
DATA COLLECTION

Subjects were asked to sit during the baseline period, indicated by a red bulb light. During this time, an event marker was sent to the dongle through the synchronous recorder. After 10 sec of baseline, the red bulb light turn off and then the bulb turned green to signal the subjects to begin walking. When subjects stop walking another event marker was sent to the dongle to mark the end of this phase.

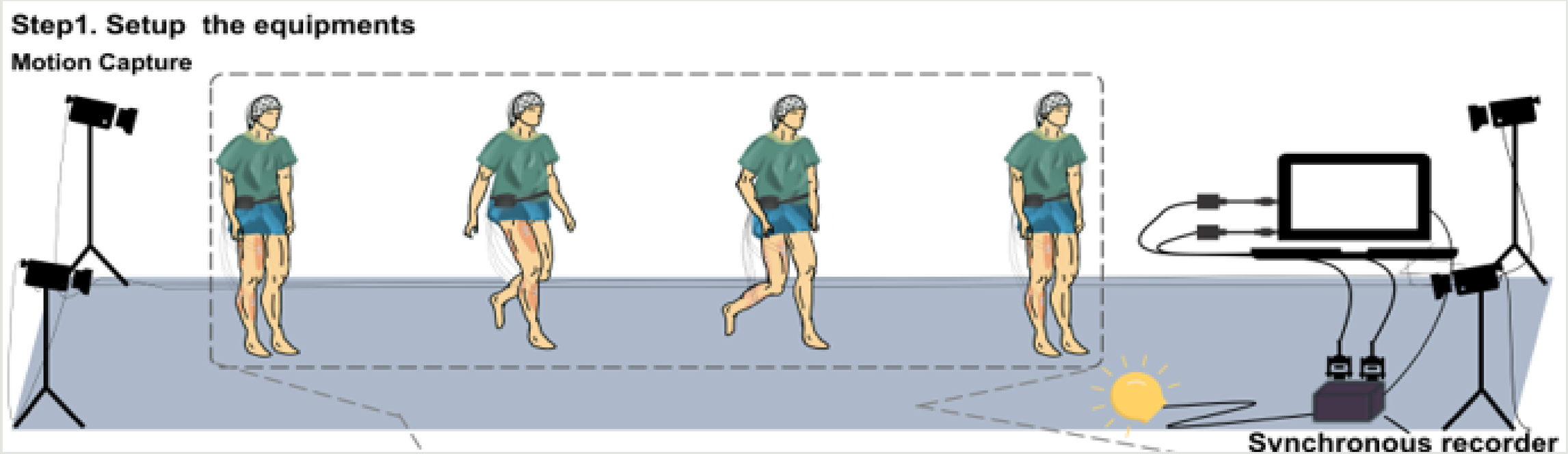
DATA COLLECTION

- 32-channel EEG cap (fs = 500)
- 8-channel surface EMG (fs =1000)
- motion capture





DATA COLLECTION



DATA COLLECTION



DATA PREPROCESSING

Electroencephalography (EEG)

Time-synchronized ⇨ Reduce mean ⇨ Detrend ⇨ Notch filter ⇨ Bandpass filter (0.5-70Hz) ⇨ Artifact subspace reconstruction (ASR) ⇨ Independent component analysis (ICA) ⇨ Z-score

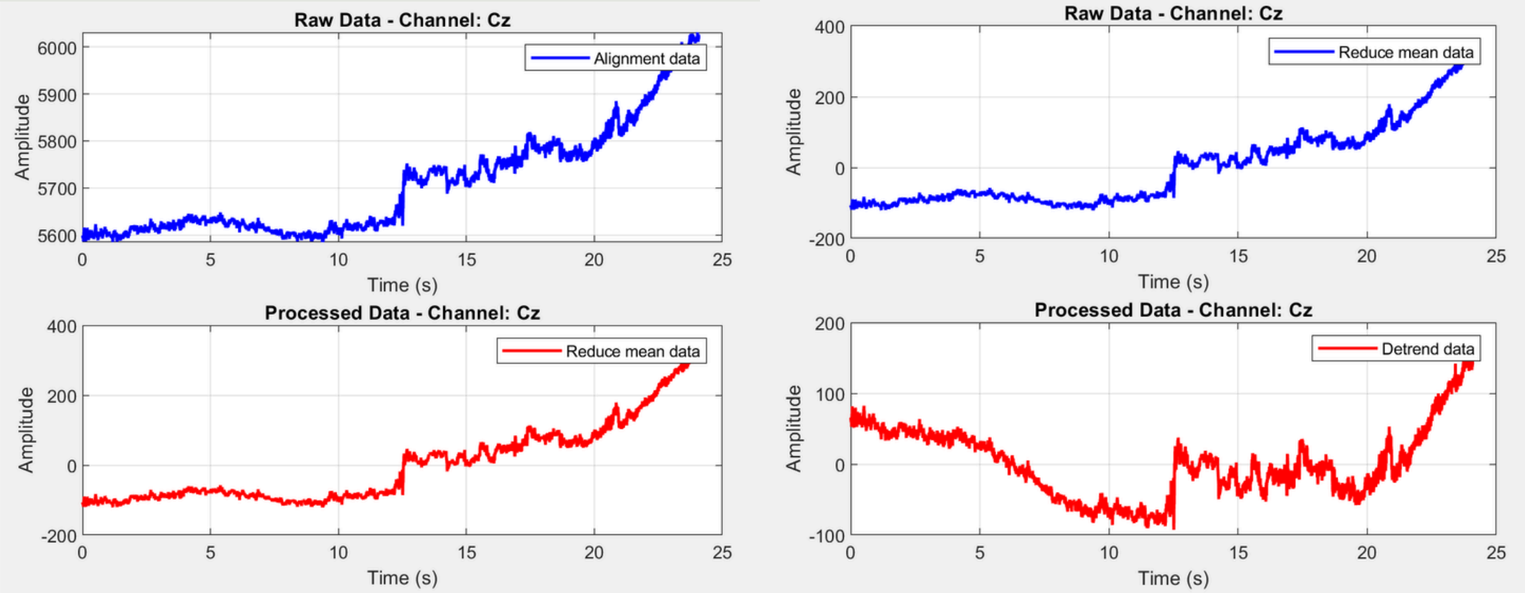
Electromyography (EMG)

Time-synchronized ⇨ Detrend ⇨ Notch filter ⇨ Frequency division (0-280Hz) ⇨ Envelope (Hilbert transform)

using EEGLAB, MATLAB

DATA PREPROCESSING-EEG

- Time-synchronized：First red light to stop walking.
- Reduce mean： Elimaniate common signal in each channel.
- Detrend： Use matlab function “detrend” to eliminate common trend in each channel.



DATA PREPROCESSING-EEG

- Notch filter： Eliminate Taiwan’s AC power 60Hz.
- Bandpass filter (0.5-70Hz)：We are interested in band frequency from 0.5Hz to 70Hz.

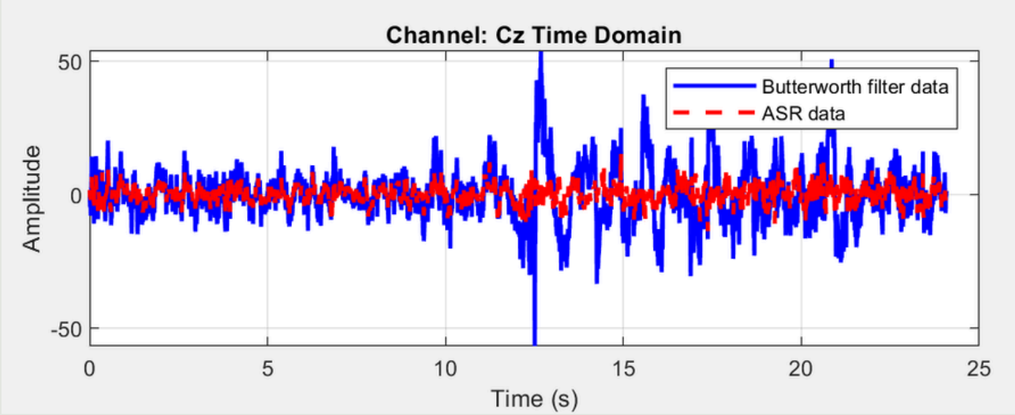
```
% 設計 notch 濾波器
fc = 60; % 濾波器中心頻率 60Hz (範圍 59-61)
[b, a] = designNotchPeakIIR('CenterFrequency', fc/(fs/2), 'Bandwidth', 2/(fs/2), 'Response', 'notch');
```

```
% 設計 Butterworth 濾波器
order = 4; % 濾波器階數
fc_low = 0.5; % 低截止頻率
fc_high = 70; % 高截止頻率
Wn = [fc_low/(fs/2), fc_high/(fs/2)]; % 歸一化截止頻率
[b, a] = butter(order, Wn, 'bandpass'); % 設計帶通濾波器
```

DATA PREPROCESSING-EEG

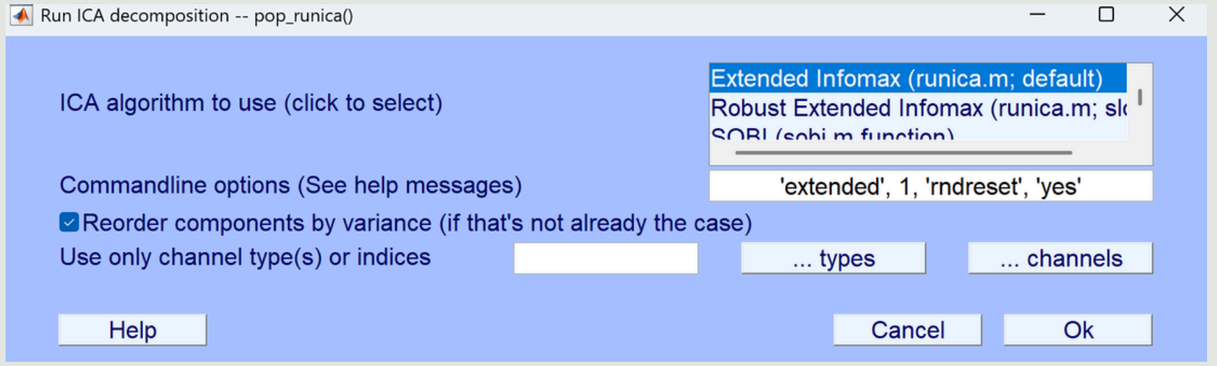
- Artifact subspace reconstruction (ASR) ： Establish a statistical model based on a short segment of clean EEG data (e.g., the first 10 seconds of the baseline period). Use this model to detect and remove high-amplitude or non-physiological artifacts from the subsequent EEG data.

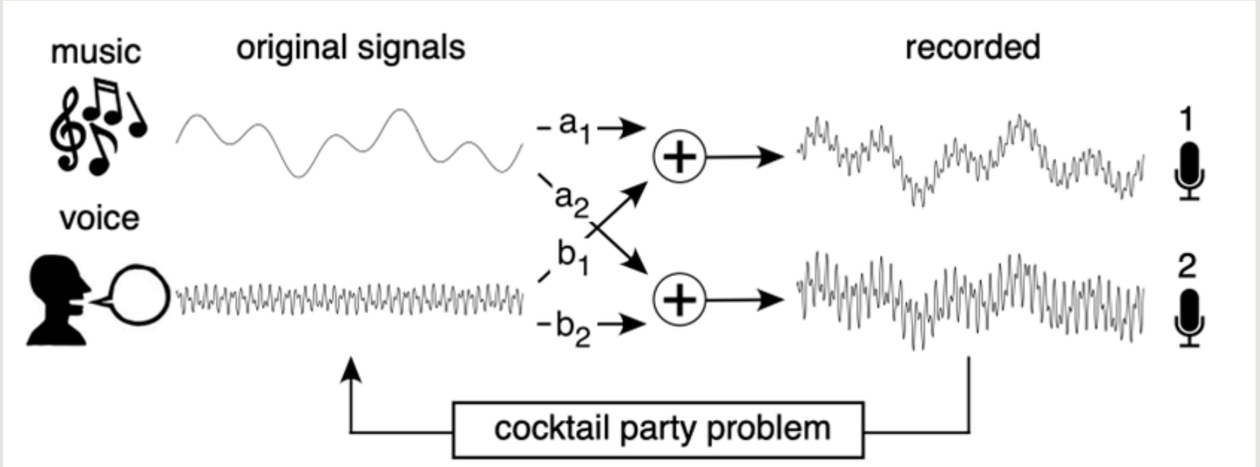
```
function [EEG, ASRbase, ASRmove] = ASR(data, sample_rate, threshold)
```



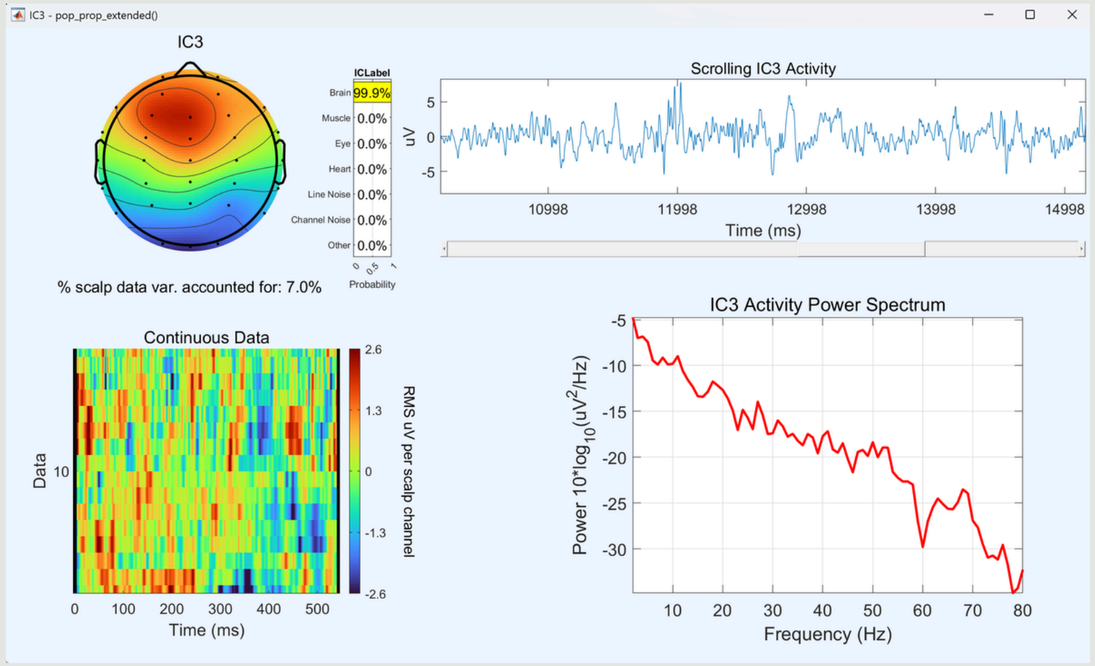
DATA PREPROCESSING-EEG

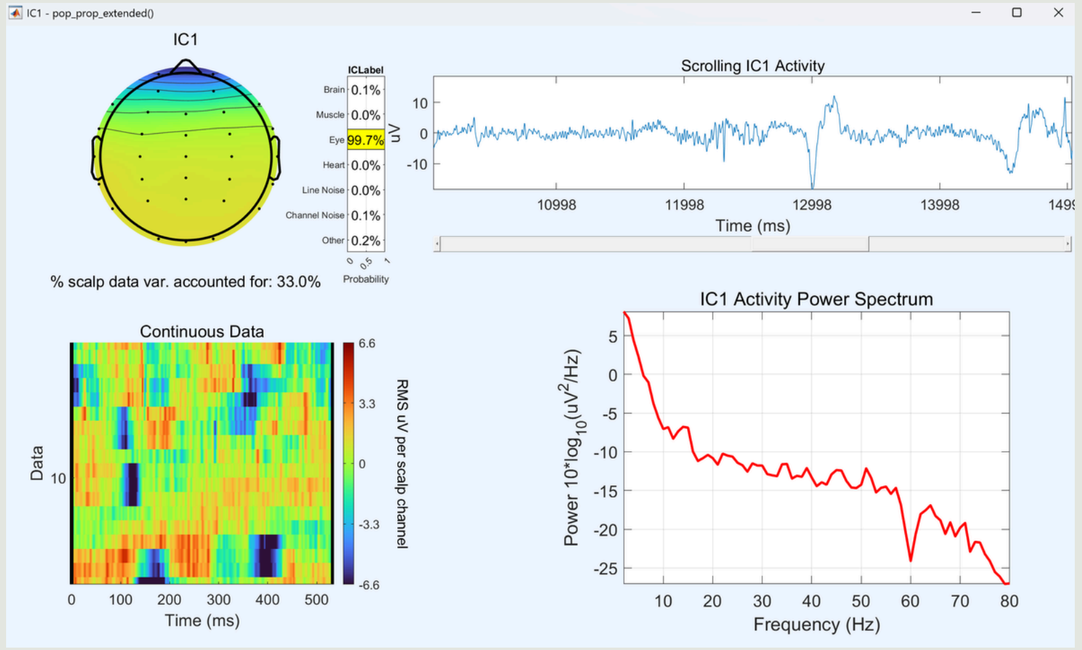
- Independent component analysis (ICA) : Separate a signal to several components with different information.





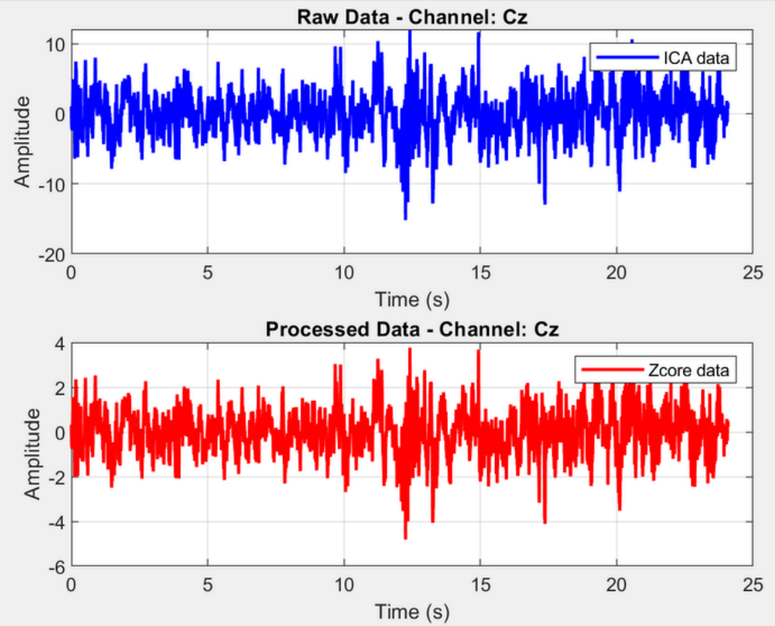
DATA PREPROCESSING-EEG





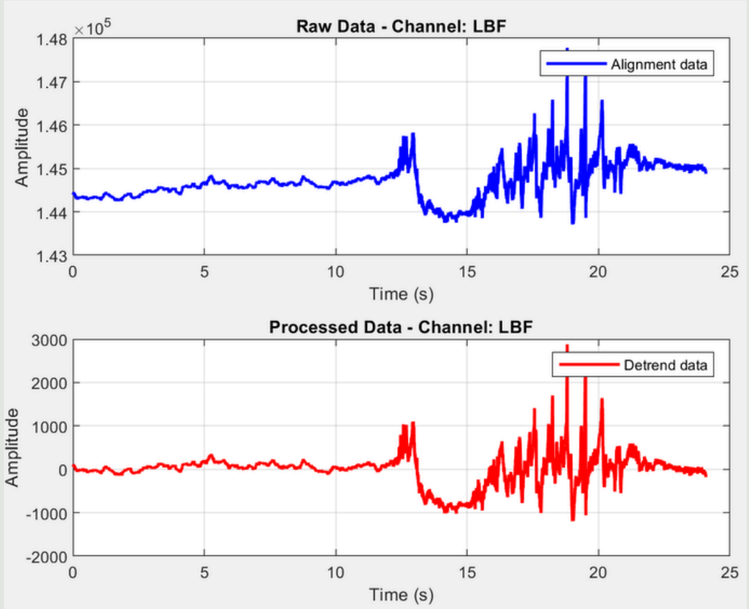
DATA PREPROCESSING-EEG

- Z-score : normalizes each EEG channel to have a mean of 0 and standard deviation of 1, allowing fair comparison between channels with different signal amplitudes.



DATA PREPROCESSING-EMG

- Time-synchronized：First red light to stop walking.
- Detrend： Use matlab function “detrend” to eliminate common trend in each channel.



DATA PREPROCESSING-EMG

- Notch filter： Eliminate Taiwan’s AC power 60Hz.
- Frequency division (0-280Hz)：We are interested in frequency band 10 - 30Hz, 30 - 45Hz, 45 - 120Hz, 135 - 280Hz.

```
% 設計 notch 濾波器
fs = 1000; % sample rate
fc = 60; % 濾波器中心頻率 60Hz (範圍 59-61)
[b, a] = designNotchPeakIIR('CenterFrequency', fc/(fs/2), 'Bandwidth', 2/(fs/2), 'Response', 'notch');
```

```
% 頻段定義（單位：Hz）
freq_bands = {
    [10, 30],
    [30, 45],
    [45, 120],
    [135, 280]
};

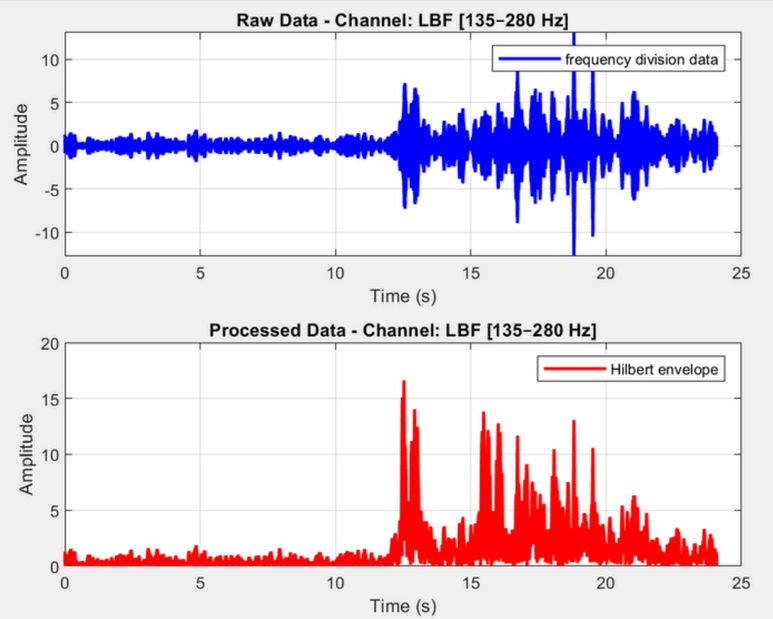
% 濾波器設定
order = 4;      % Butterworth 濾波器階數
fs = 1000;      % 取樣率
```

```
% 計算歸一化截止頻率
fc_low = freq_bands{b}(1);
fc_high = freq_bands{b}(2);
Wn = [fc_low, fc_high] / (fs / 2);

% 設計 Butterworth 濾波器
[b_butter, a_butter] = butter(order, Wn, 'bandpass');
```

DATA PREPROCESSING-EMG

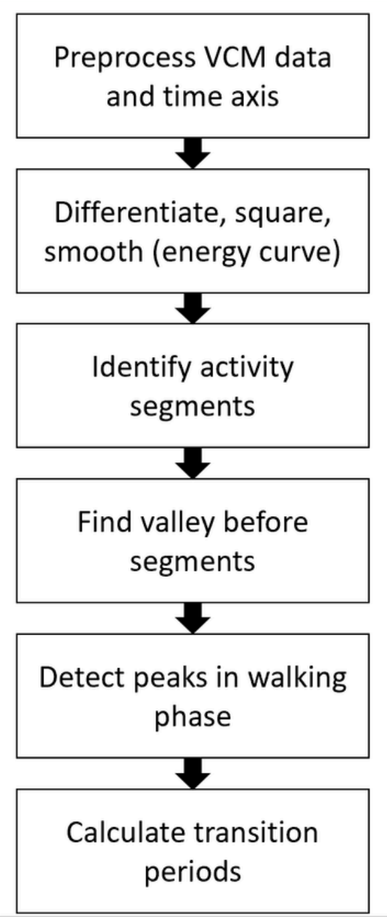
- Envelope (Hilbert transform) ：convert the original high-frequency oscillatory signal into a smooth and easily interpretable amplitude trend.

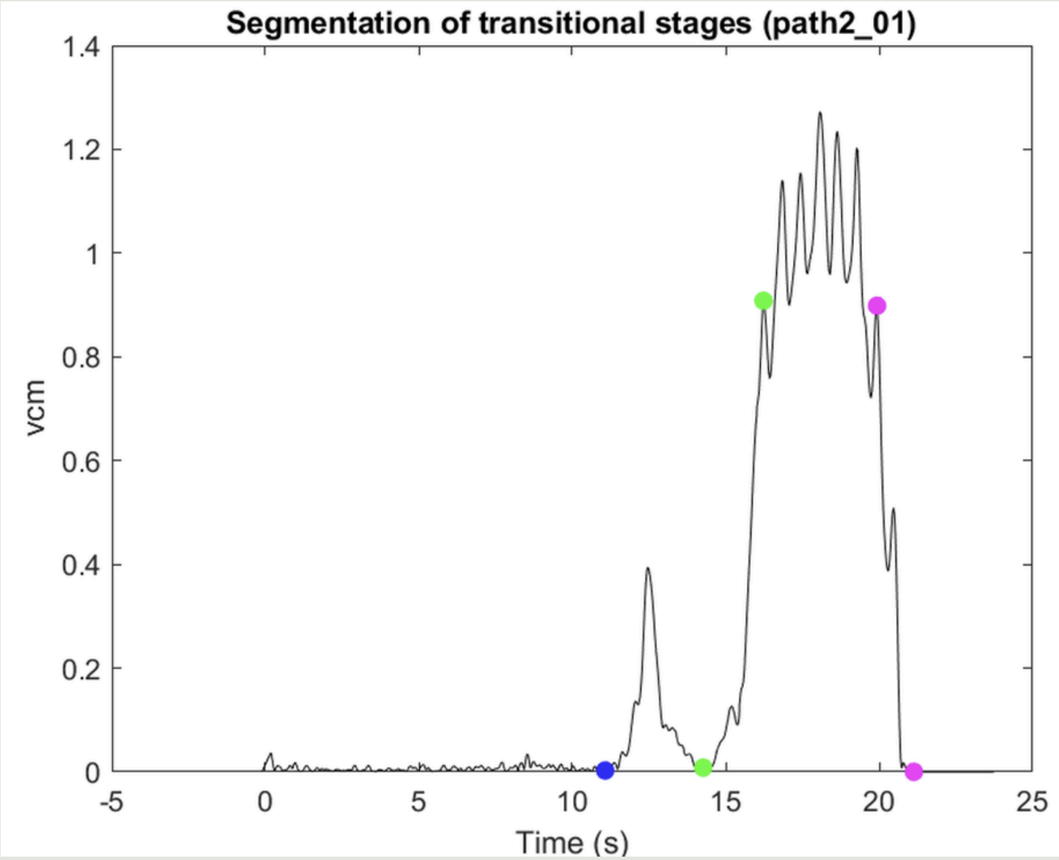



```
% Hilbert transform（針對每個 channel）
analytic_signal = hilbert(band_data')';
envelope_data = abs(analytic_signal);
```

SEGMENTATION OF TRANSITIONAL STAGES

- Using the velocity of the center of mass (VCM) from mocap and pan tompkins algorithm to define transitional stages(Sit-to-Stand, Stand-to-Walk, Walk-to-Stop).

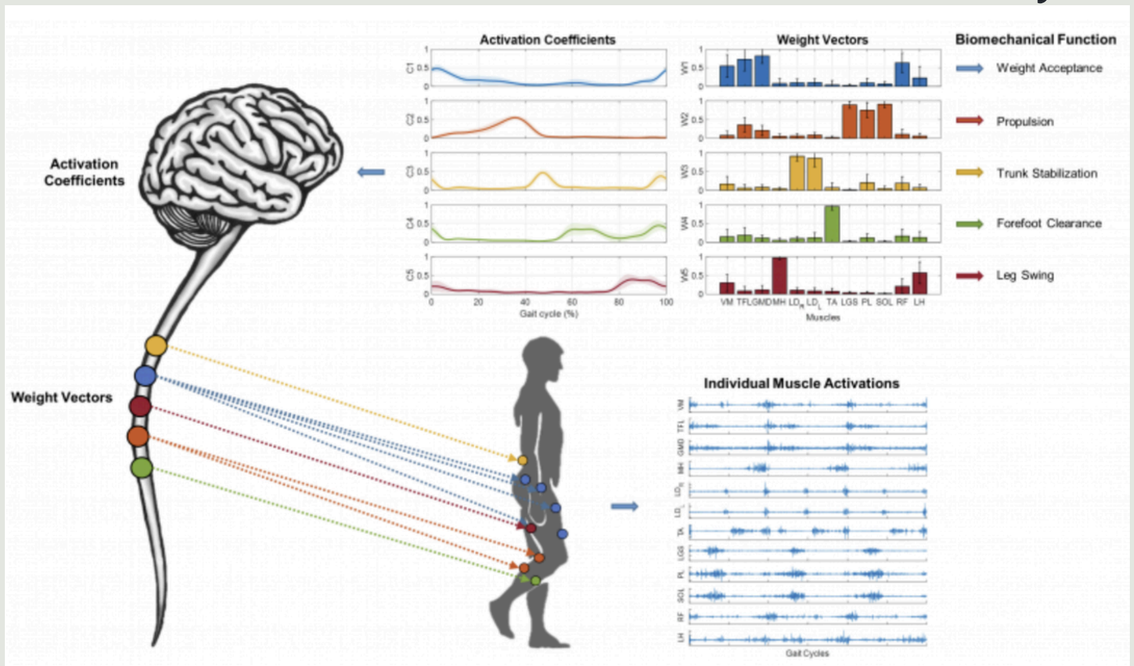




MUSCLE SYNERGY

Definition : A group of muscles that are coordinated and activated together to produce a specific movement or action.

The brain and spinal cord are believed to simplify the control of muscles by grouping them into functional units called muscle synergies.



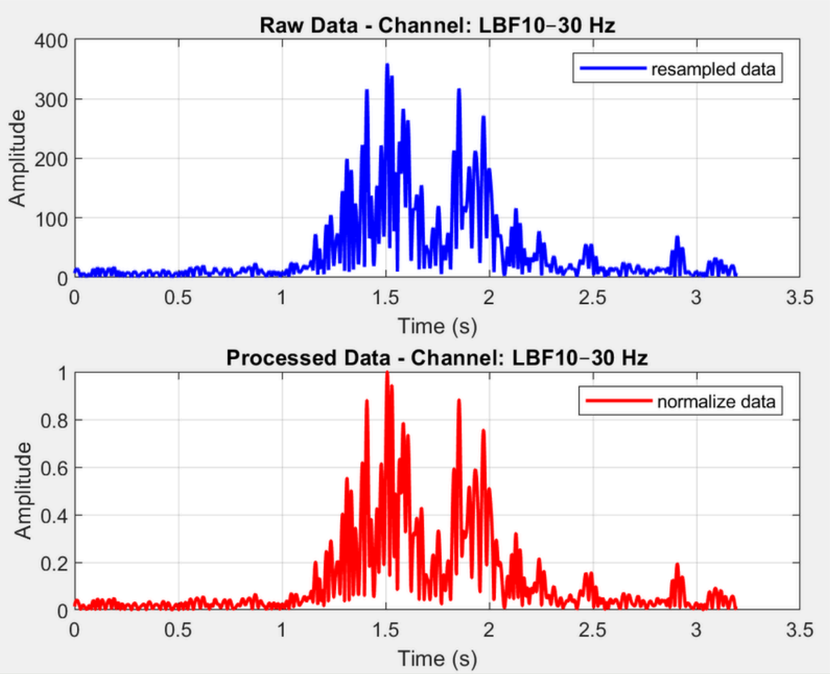
MUSCLE SYNERGY

EMG preprocessed signal after segmentation of transitional stages ⇨ Resample (sample rate same as EEG) ⇨ Normalize ⇨ Non-negative matrix factorization (NMF) ⇨ Classify muscle synergy

MUSCLE SYNERGY

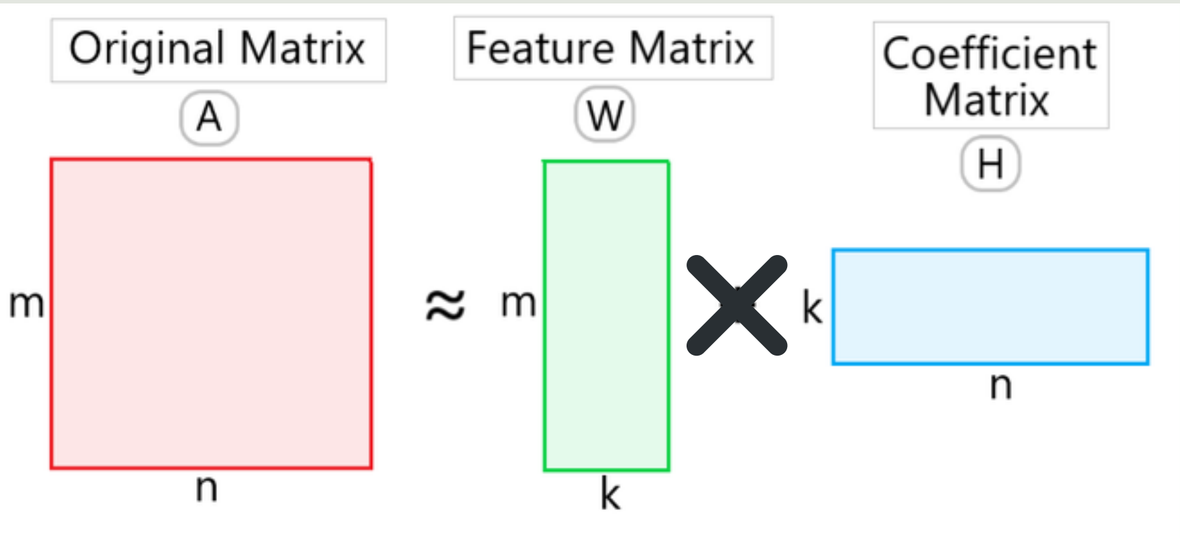
- Resample： Using matlab function “resample” to let EEG and EMG have the same sample rate (fs = 500) for further analysis
- Normalize： Using min-max normalize, prepare for Non-negative matrix factorization.

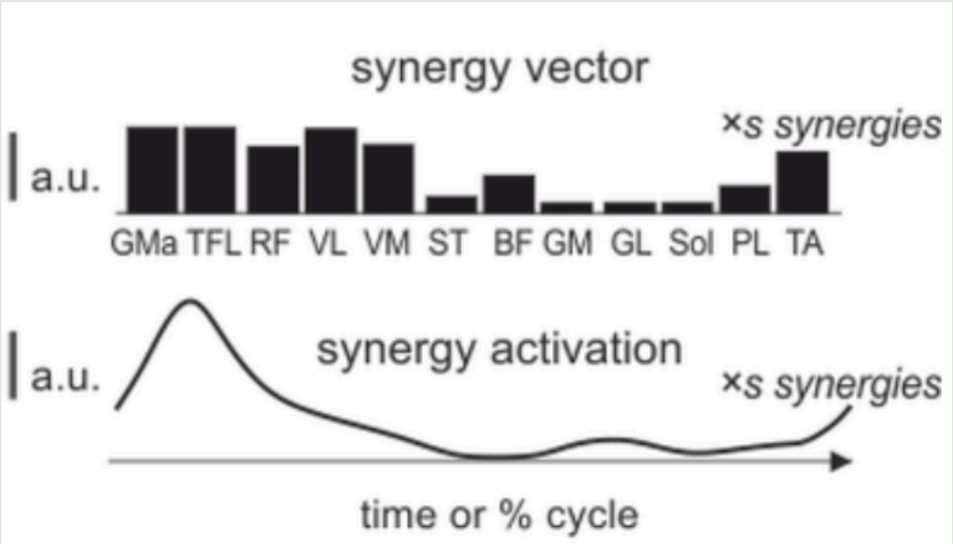
$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$



MUSCLE SYNERGY

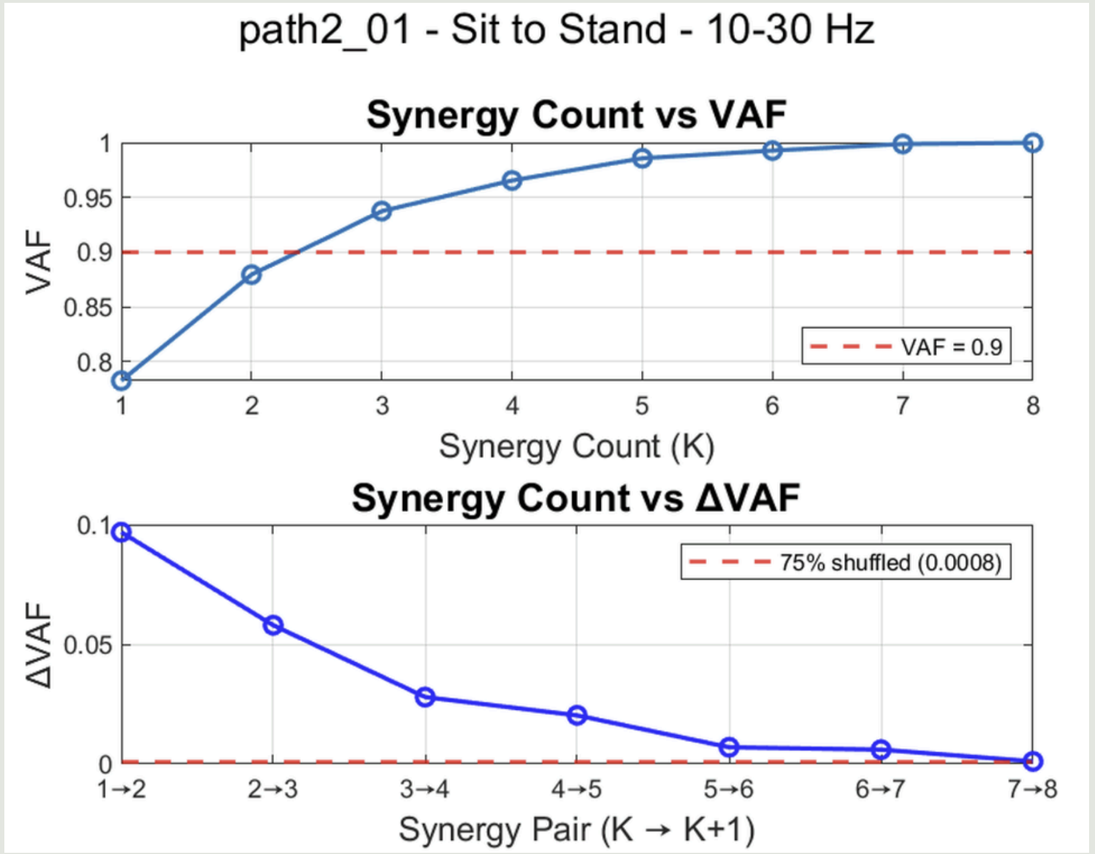
- Non-negative matrix factorization (NMF) : Using toolbox "The NMF MATLAB Toolbox". Number of synergies were determined by the Variance account for (VAF). I used (1) VAF ≥ 0.9 and (2) detla VAF ≥ 0.75 for shuffle data as criteria.





MUSCLE SYNERGY

```
% 計算 VAF
SST = sum(sum((data - mean(data, 'all')).^2));
SSE = sum(sum((data - reconstructed).^2));
VAF_per_k(k_idx) = 1 - (SSE / SST);
```



MUSCLE SYNERGY

- Classify muscle synergy：Integrate the EMG synergy results from multiple subjects, identify common synergy patterns through clustering and matching.

```
% 執行 k-means clustering 找共識 synergy
[cluster_idx, cluster_centers] = kmeans(all_synergies, common_K,
    'Distance', 'cosine', 'Replicates', 20, 'MaxIter', 10000);
```

```
% 配對
A_t = A';  % K × muscles
dist_tmp = pdist2(A_t, cluster_centers, 'cosine');  % K × K
matching = matchpairs(dist_tmp, 1);  % K x 2
```

MUSCLE SYNERGY



SIGNAL PROCESSING PIPELINE

Electroencephalography (EEG)

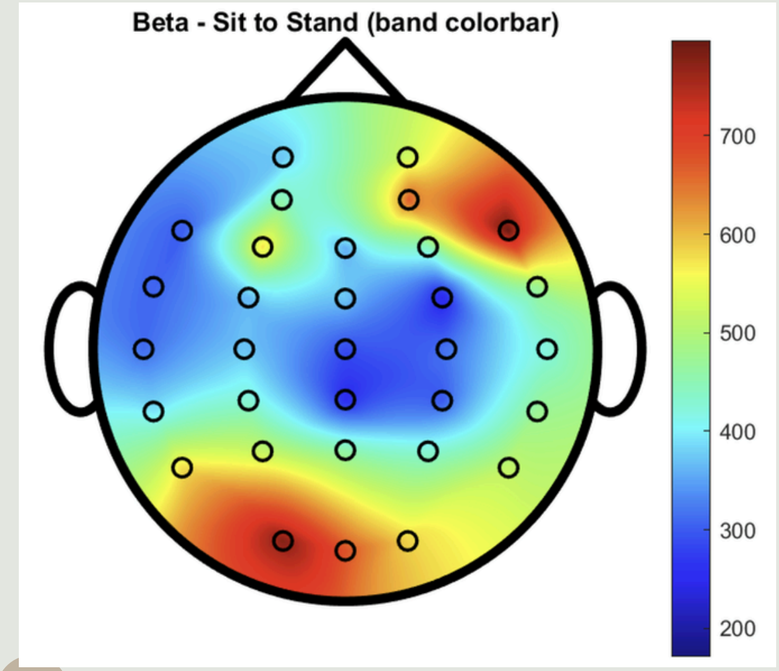
Time-synchronized ⇨ Reduce mean ⇨ Detrend ⇨ Notch filter ⇨ Bandpass filter (0.5-70Hz) ⇨ Artifact subspace reconstruction (ASR) ⇨ Independent component analysis (ICA) ⇨ Z-score ⇨ Segmentation of transitional stages

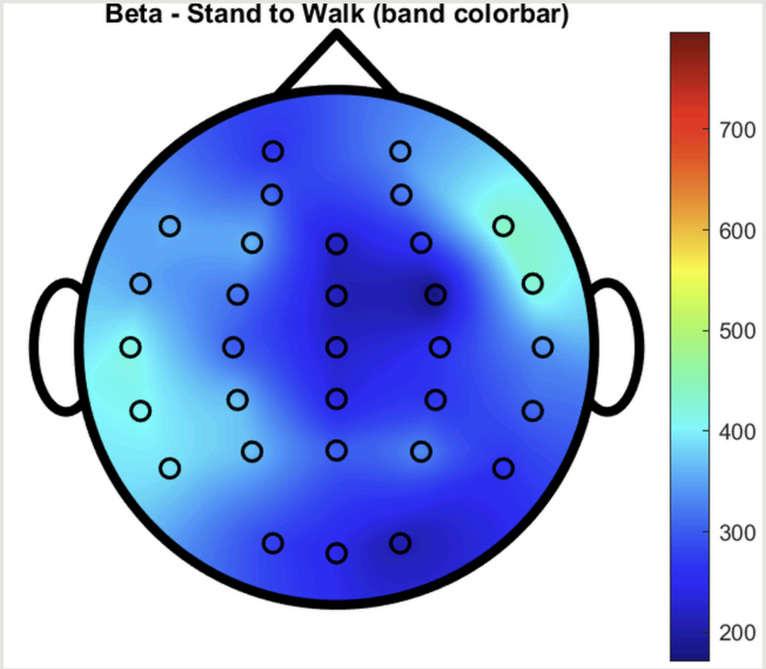
Electromyography (EMG)

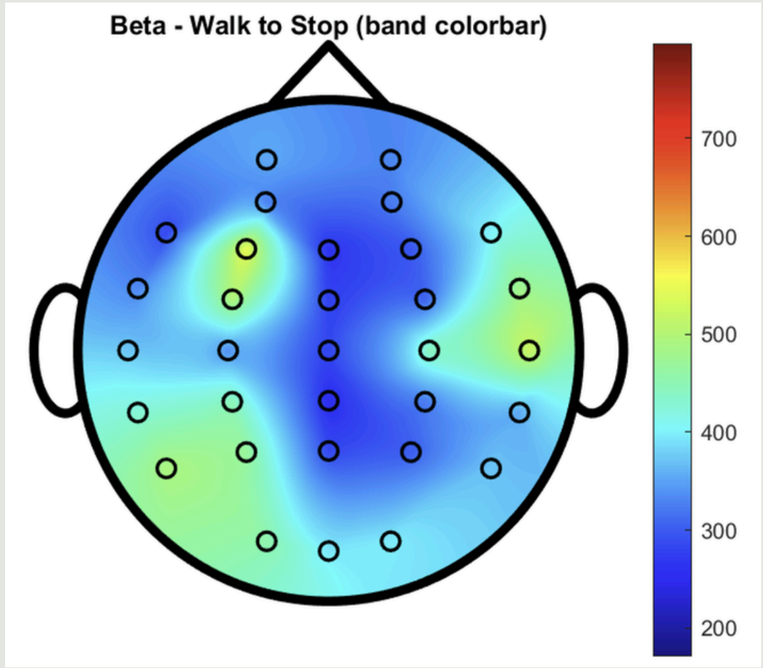
Time-synchronized ⇨ Detrend ⇨ Notch filter ⇨ Frequency division (0-280Hz) ⇨ Envelope (Hilbert transform) ⇨ Segmentation of transitional stages ⇨ Resample ⇨ Normalize ⇨ Non-negative matrix factorization(NMF) ⇨ Classify muscle synergy

RESULTS-EEG

beta band 13-30Hz in different transitional stages

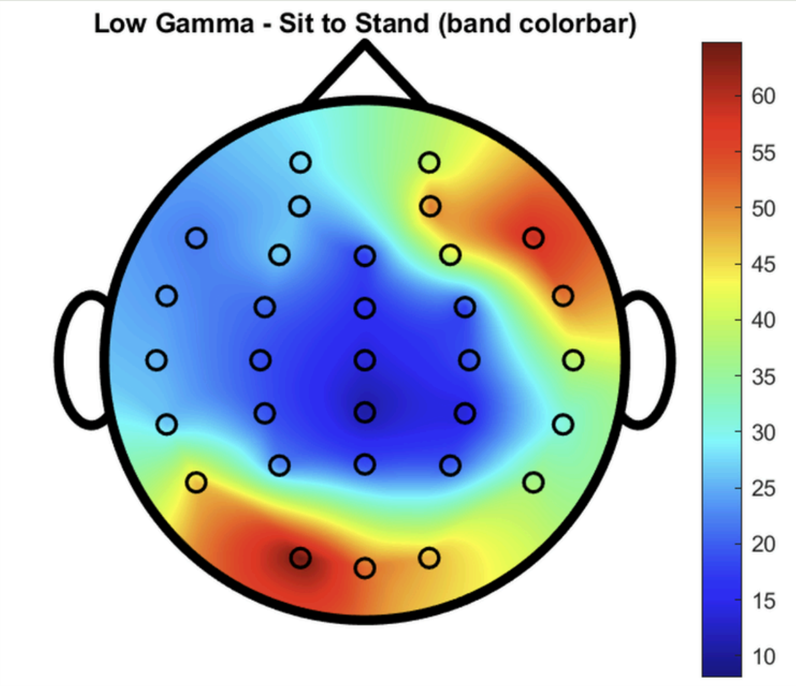


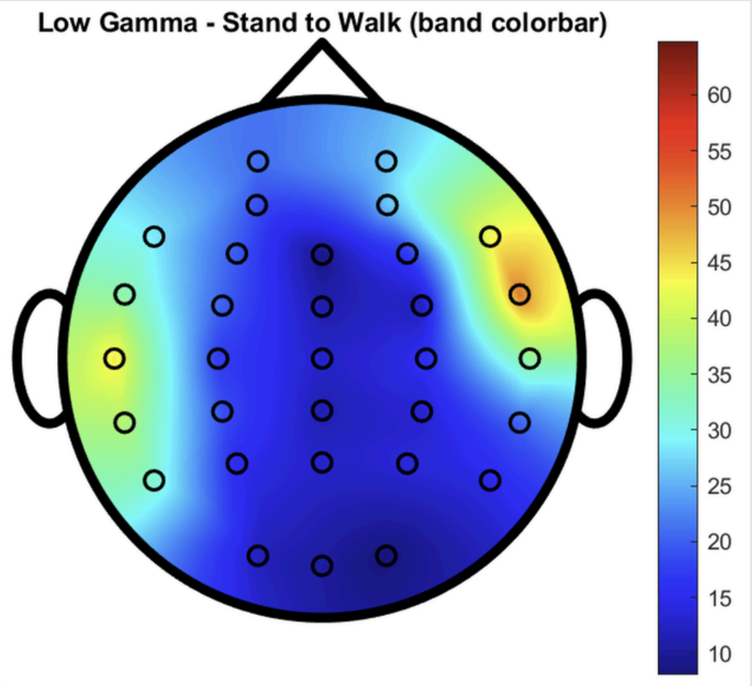


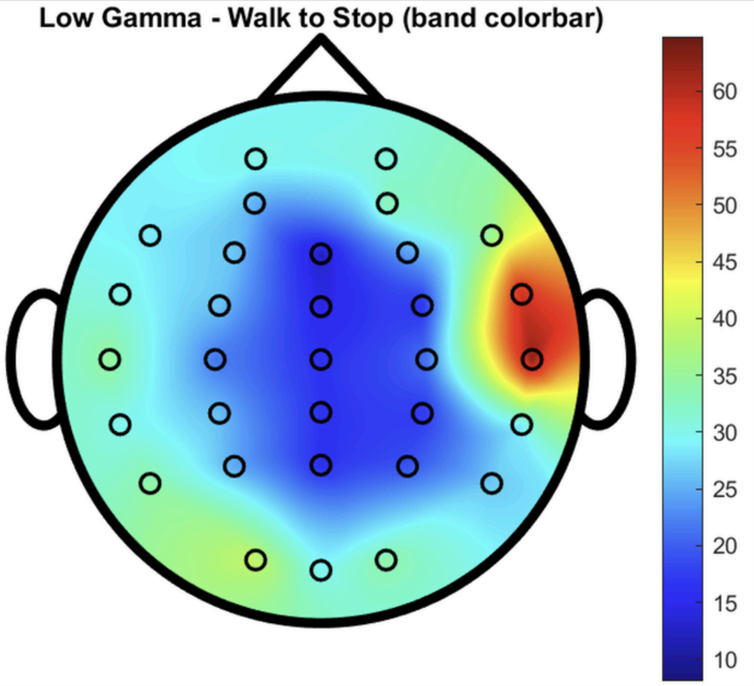


RESULTS-EEG

low gamma band 32-46Hz in different transitional stages

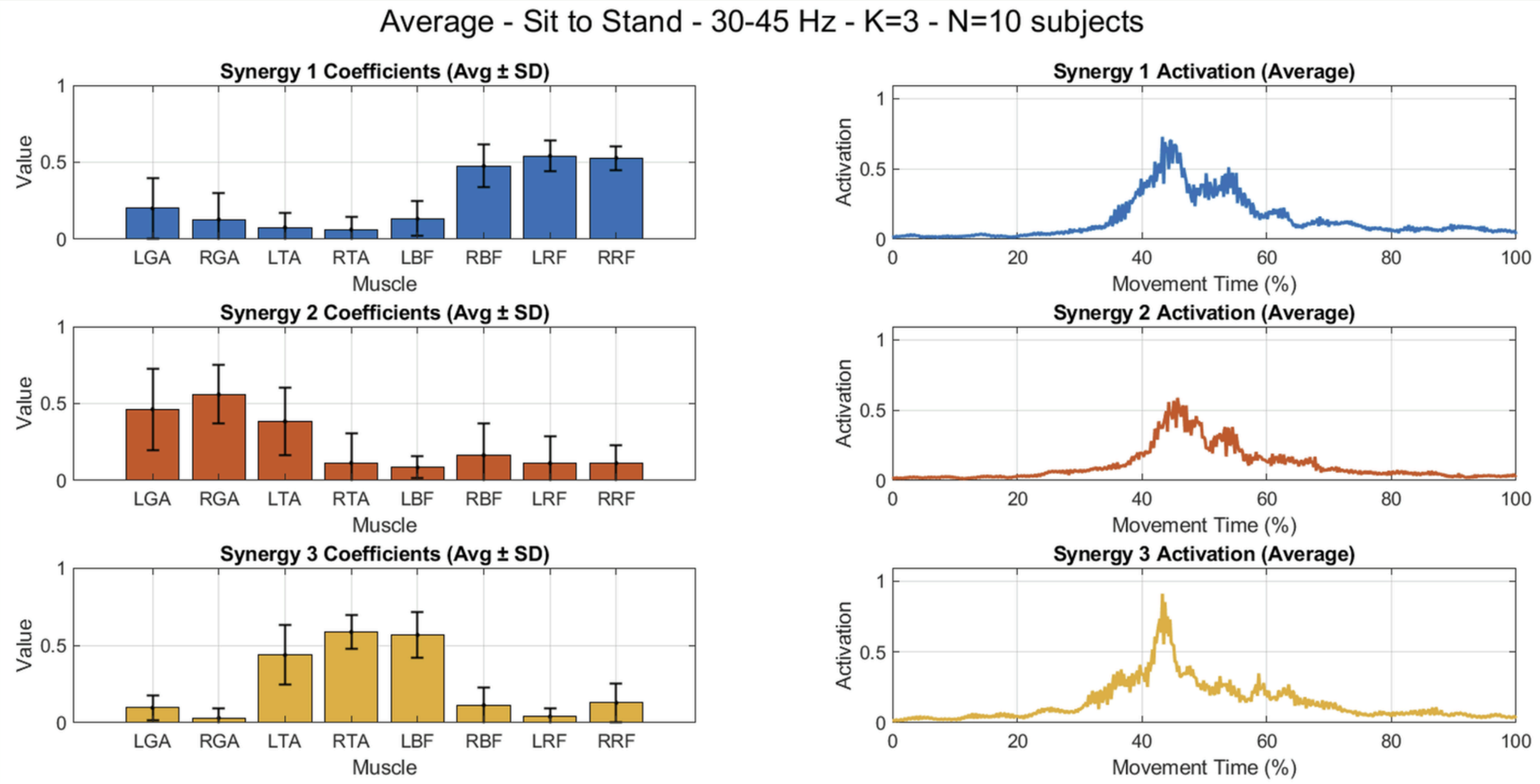






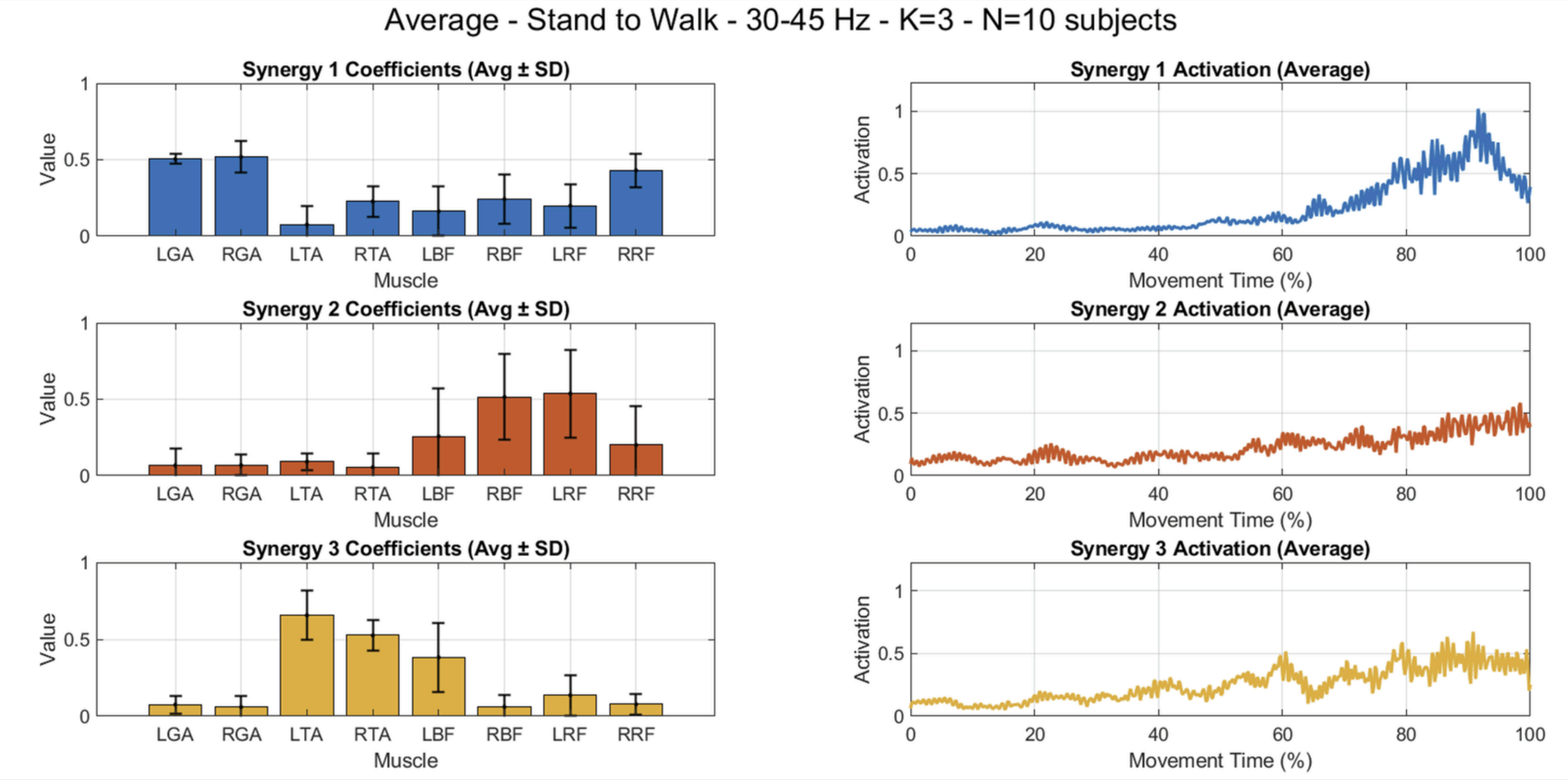
RESULTS-MUSCLE SYNERGY

frequency band 30-45Hz in Sit-to-Stand stages



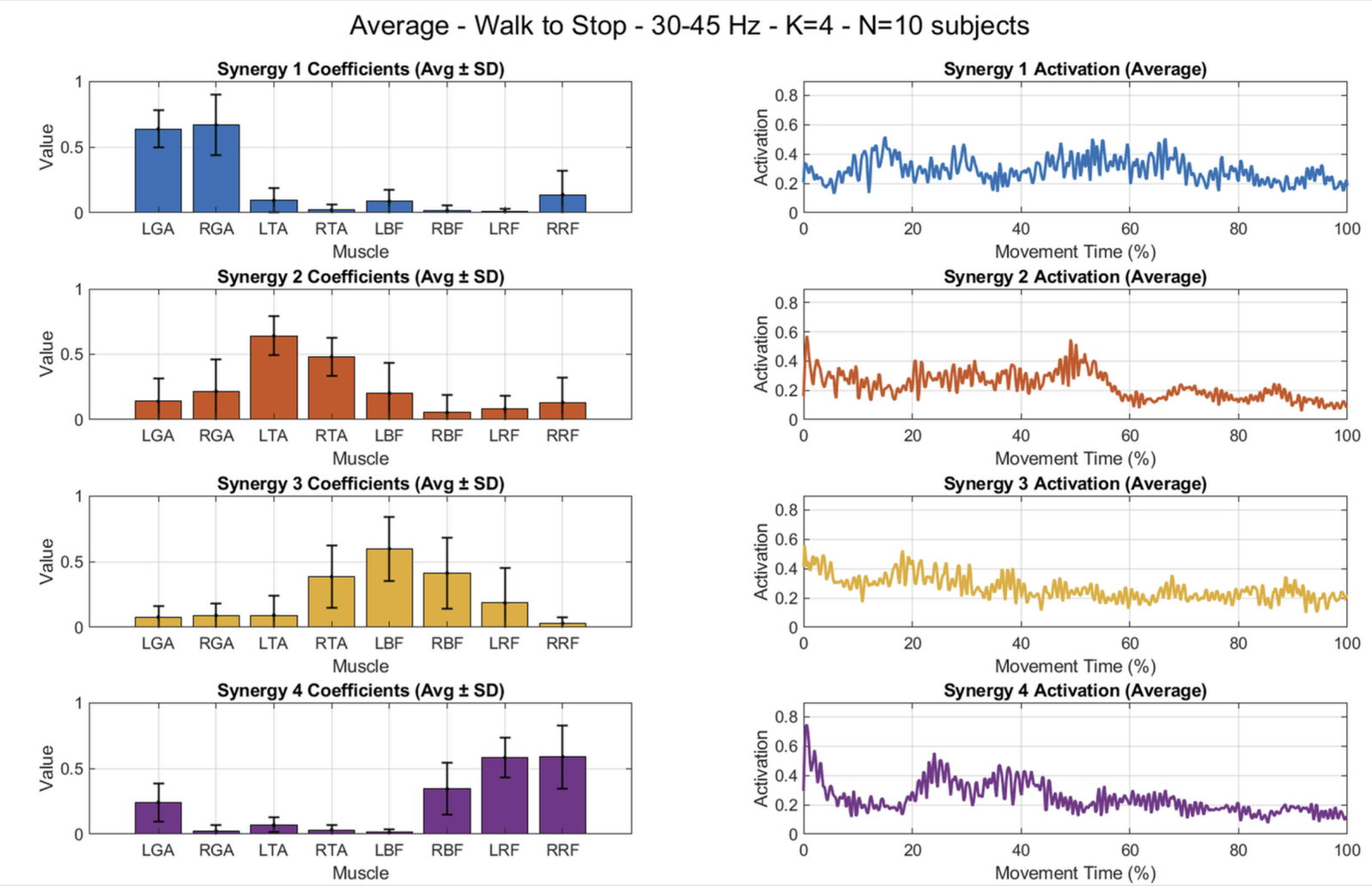
RESULTS-MUSCLE SYNERGY

frequency band 30-45Hz in Stand-to-Walk stages



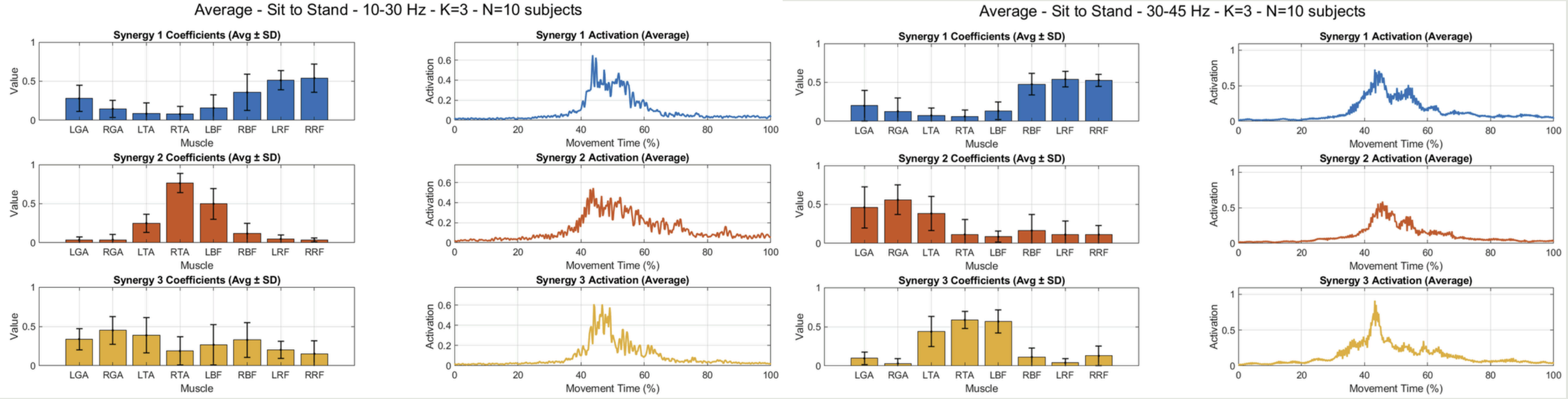
RESULTS-MUSCLE SYNERGY

frequency band 30-45Hz in Walk-to-Stop stages



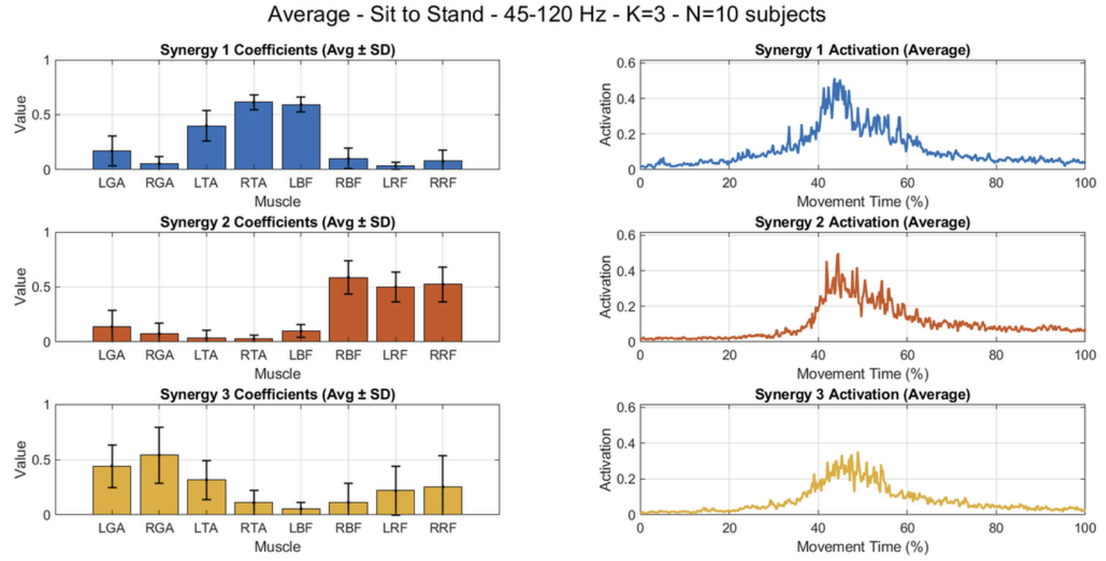
RESULTS-MUSCLE SYNERGY

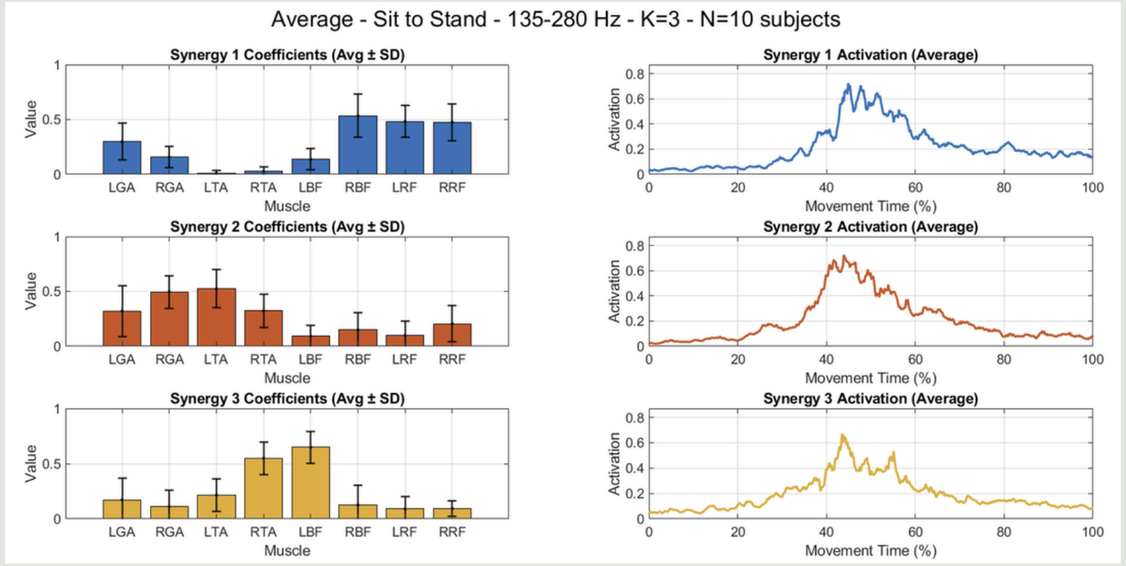
different frequency bands in Sit-to-Stand stage



RESULTS-MUSCLE SYNERGY

different frequency bands in Sit-to-Stand stage





CONCLUSION

Electroencephalography (EEG)

During the Sit-to-Stand stage, increased beta and low gamma power was observed in the right frontal and left occipital regions, suggesting the brain's involvement in movement preparation and postural control during transitional movements.

Electromyography (EMG)

Two consistent muscle synergies were identified during transitional movements:

1. LTA, RTA, and LBF emerged in both Sit-to-Stand and Stand-to-Walk stages within the 30–45 Hz band.
2. RBF, LRF, and RRF were recruited across all frequency bands during Sit-to-Stand.

These findings suggest a central strategy to engage stable muscle modules during dynamic transitions

REFERENCES

European Journal of Applied Physiology (2021) 121:1009–1025
https://doi.org/10.1007/s00421-021-04604-9
How to improve the muscle synergy analysis methodology?

https://www.sciencedirect.com/science/article/abs/pii/S1746809424005135
A novel method for movement quality analysis of lower limb joints using surface electromyography signals and k-means clustering technique

https://www.nature.com/articles/s41598-023-43148-0
Sit-to-walk strategy classification in healthy adults using hip and knee joint angles at gait initiation

Thank You

For your attention